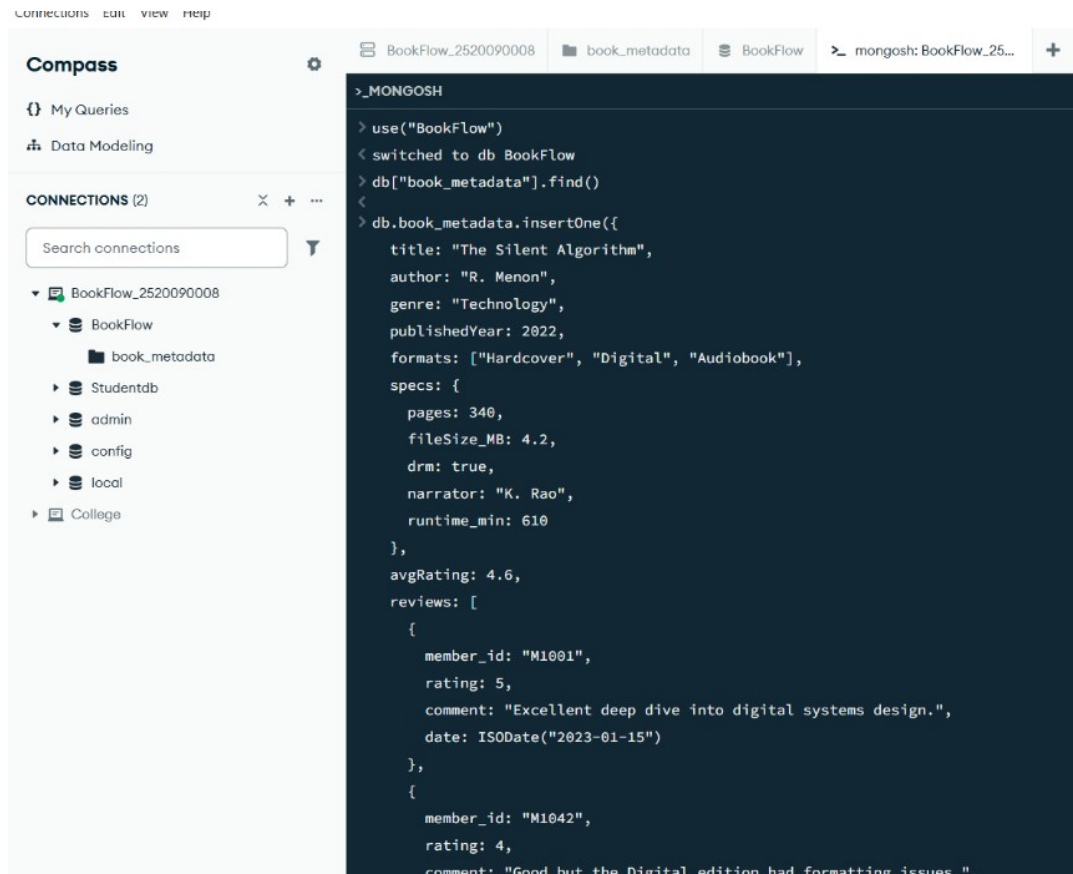


WEEK-3 Practical

2520090071-Vignesh Kumar

Task-1:

Inserting the First Book Document



Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- BookFlow_2520090008
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
- College

```
>_MONGOSH

formats: ["Hardcover", "Digital", "Audiobook"],
specs: {
  pages: 340,
  fileSize_MB: 4.2,
  drm: true,
  narrator: "K. Rao",
  runtime_min: 610
},
avgRating: 4.6,
reviews: [
  {
    member_id: "M1001",
    rating: 5,
    comment: "Excellent deep dive into digital systems design.",
    date: ISODate("2023-01-15")
  },
  {
    member_id: "M1042",
    rating: 4,
    comment: "Good but the Digital edition had formatting issues.",
    date: ISODate("2023-02-20")
  }
]
});
< {
  acknowledged: true,
  insertedId: ObjectId('6a8d23960e5a8db208c7beee')
}
```

Inserting Multiple Book Documents

Compass

My Queries

Data Modeling

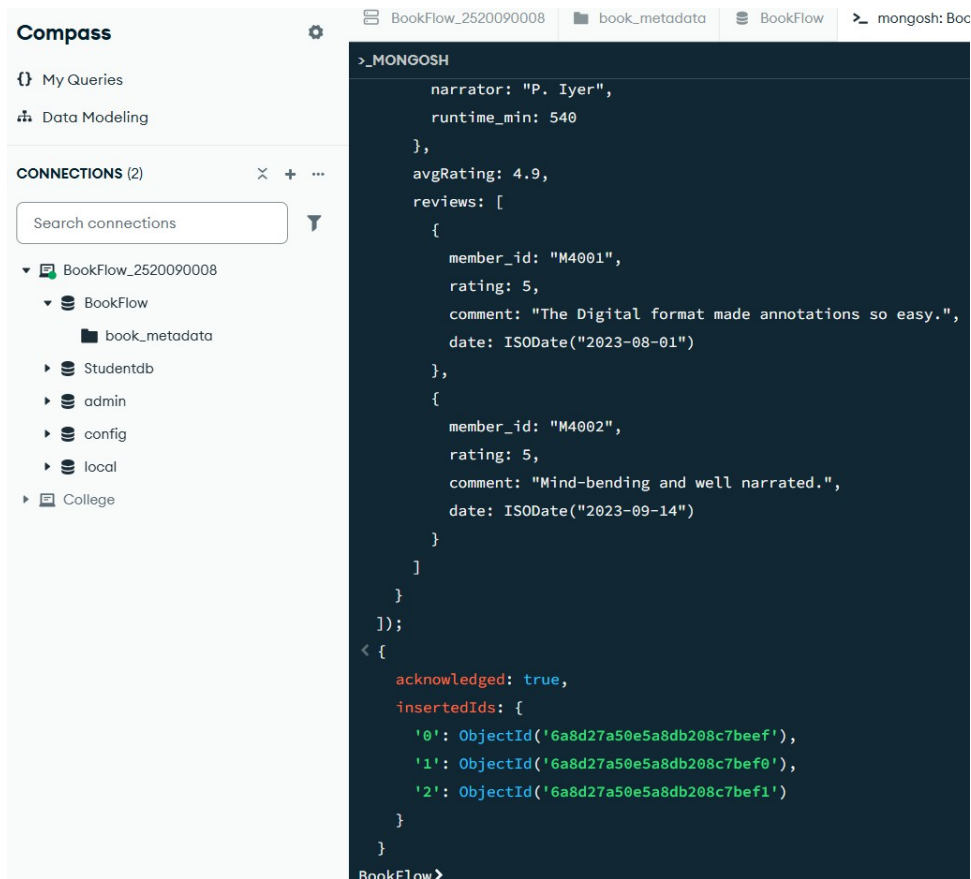
CONNECTIONS (2)

Search connections

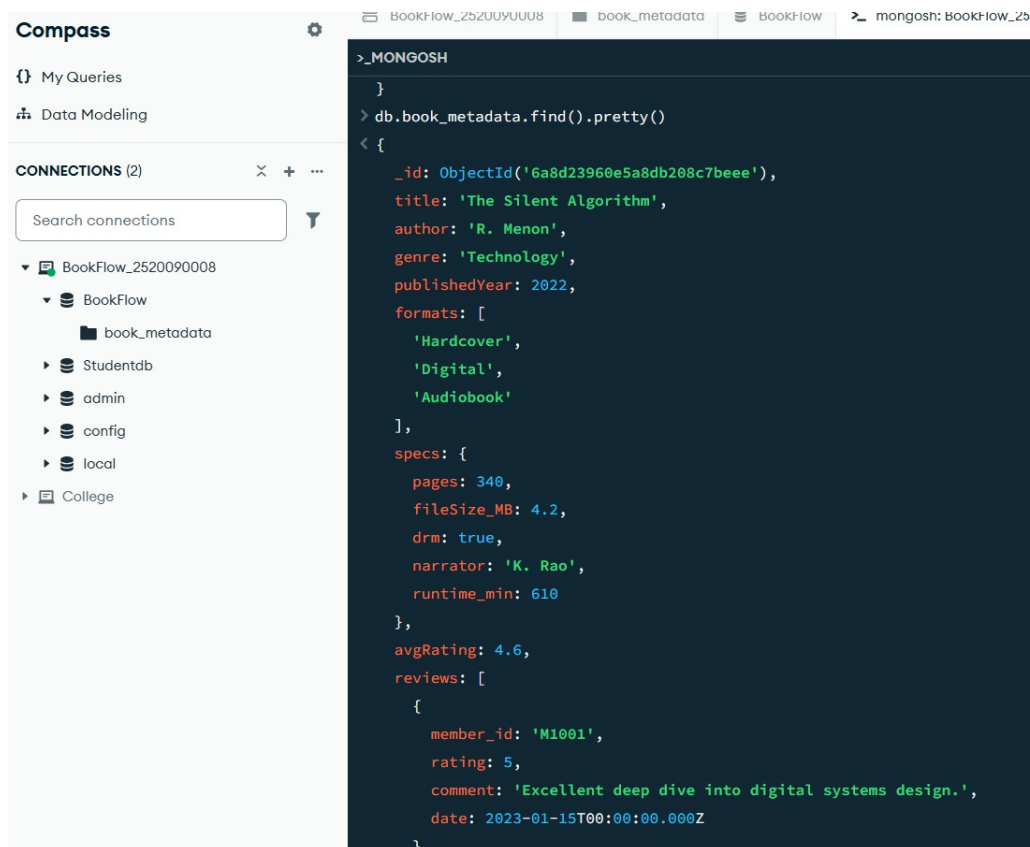
- BookFlow_2520090008
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
- College

```
>_MONGOSH

}
> db.book_metadata.insertMany([
  {
    title: "Ocean of Code",
    author: "A. Fernandes",
    genre: "Technology",
    publishedYear: 2021,
    formats: ["Digital", "Paperback"],
    specs: {
      pages: 210,
      fileSize_MB: 2.1,
      drm: false
    },
    avgRating: 4.2,
    reviews: [
      {
        member_id: "M2001",
        rating: 4,
        comment: "Solid Digital read, clean layout.",
        date: ISODate("2022-05-10")
      },
      {
        member_id: "M2002",
        rating: 5,
        comment: "Best programming book this year.",
        date: ISODate("2022-06-01")
      }
    ]
  }
])
},
```

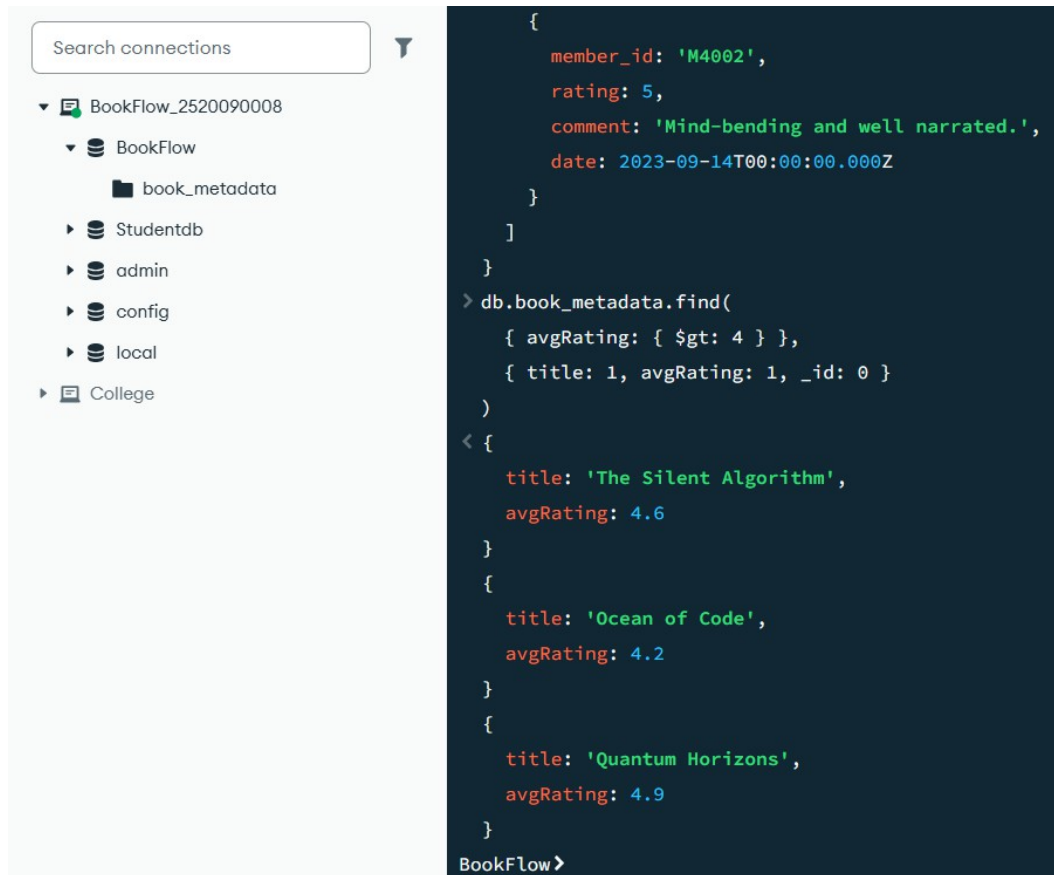


Displaying the Book Metadata Collection



TASK - 2

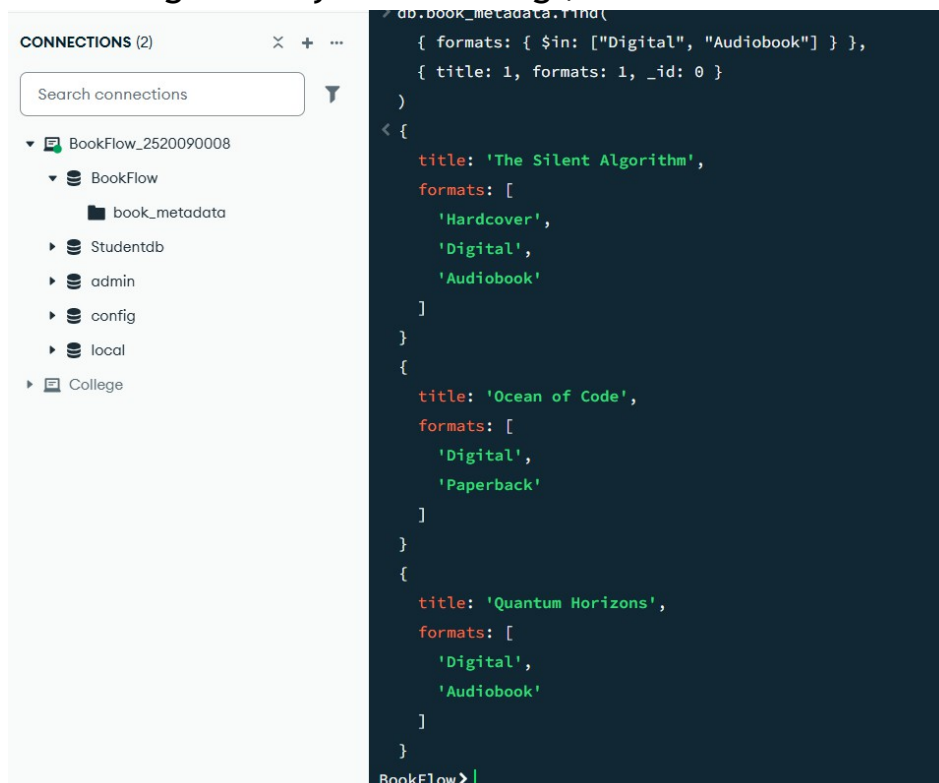
1. Finding Books Rated Above 4 Using \$gt



The screenshot shows a MongoDB interface with a sidebar on the left and a main panel on the right. The sidebar, titled "Search connections", lists several databases: BookFlow_2520090008, BookFlow, book_metadata, Studentdb, admin, config, local, and College. The main panel displays a JSON document representing a book entry. The document has fields: member_id: 'M4002', rating: 5, comment: 'Mind-bending and well narrated.', and date: 2023-09-14T00:00:00.000Z. Below the document, a query is shown: > db.book_metadata.find({ avgRating: { \$gt: 4 } }, { title: 1, avgRating: 1, _id: 0 }). The result shows three books: 'The Silent Algorithm' with an avgRating of 4.6, 'Ocean of Code' with an avgRating of 4.2, and 'Quantum Horizons' with an avgRating of 4.9. The prompt "BookFlow>" is visible at the bottom of the main panel.

```
{
  member_id: 'M4002',
  rating: 5,
  comment: 'Mind-bending and well narrated.',
  date: 2023-09-14T00:00:00.000Z
}
]
}
> db.book_metadata.find(
  { avgRating: { $gt: 4 } },
  { title: 1, avgRating: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  avgRating: 4.9
}
BookFlow>
```

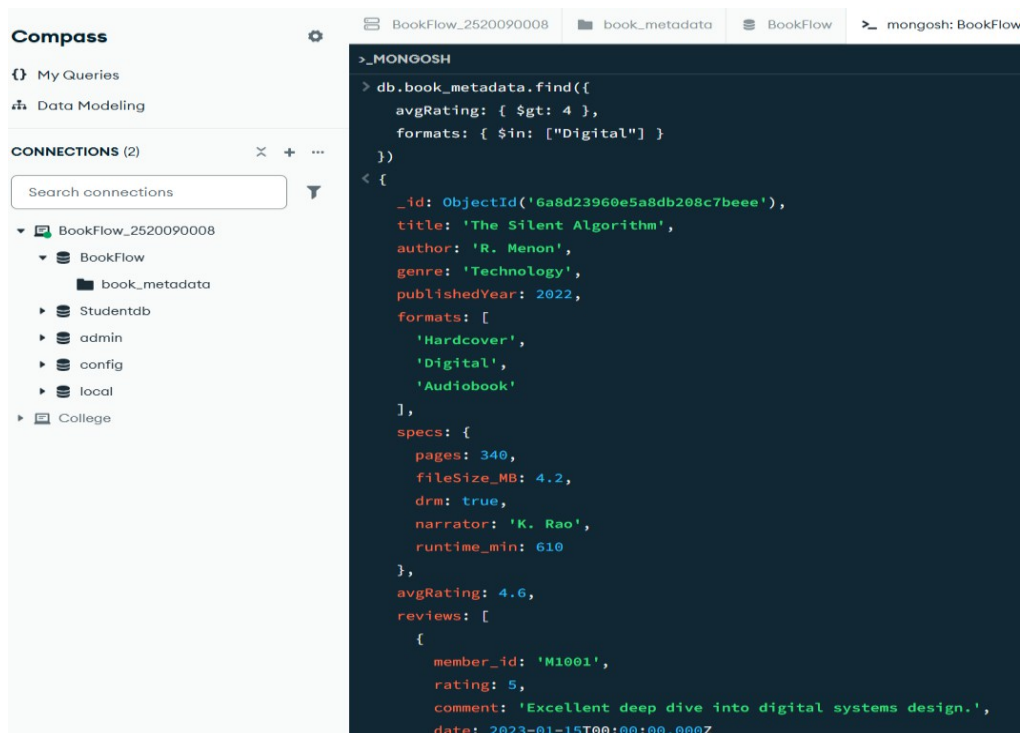
2. Finding Books by Format Using \$in



The screenshot shows a MongoDB interface with a sidebar on the left and a main panel on the right. The sidebar, titled "Search connections", lists several databases: BookFlow_2520090008, BookFlow, book_metadata, Studentdb, admin, config, local, and College. The main panel displays a JSON document representing a book entry. The document has fields: title: 'The Silent Algorithm', formats: ['Hardcover', 'Digital', 'Audiobook'], and avgRating: 4.6. Below the document, a query is shown: > db.book_metadata.find({ formats: { \$in: ['Digital', 'Audiobook'] } }, { title: 1, formats: 1, _id: 0 }). The result shows three books: 'The Silent Algorithm' with formats ['Hardcover', 'Digital', 'Audiobook'], 'Ocean of Code' with formats ['Digital', 'Paperback'], and 'Quantum Horizons' with formats ['Digital', 'Audiobook']. The prompt "BookFlow>" is visible at the bottom of the main panel.

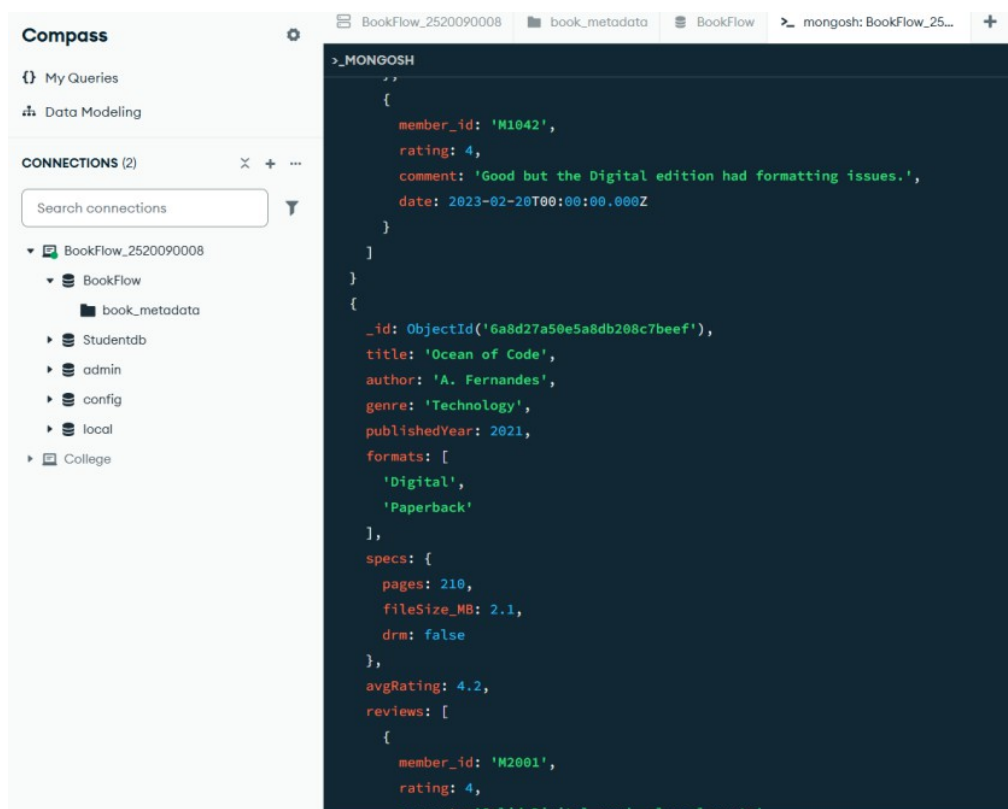
```
> db.book_metadata.find(
  { formats: { $in: ['Digital', 'Audiobook'] } },
  { title: 1, formats: 1, _id: 0 }
)
< {
  title: 'The Silent Algorithm',
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  avgRating: 4.6
}
{
  title: 'Ocean of Code',
  formats: [
    'Digital',
    'Paperback'
  ],
  avgRating: 4.2
}
{
  title: 'Quantum Horizons',
  formats: [
    'Digital',
    'Audiobook'
  ],
  avgRating: 4.9
}
BookFlow>
```

3. Combining Rating and Format Conditions



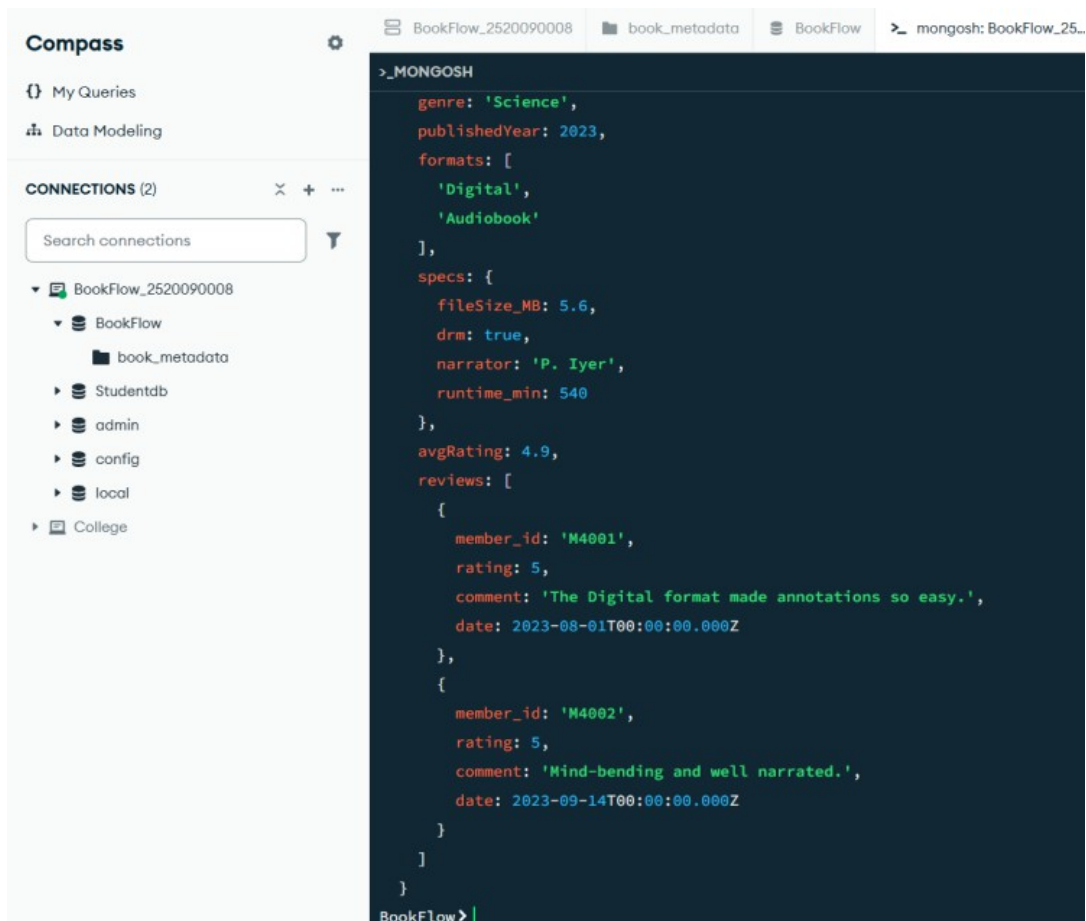
The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists the database 'BookFlow_2520090008' and its collections 'BookFlow' and 'book_metadata'. The main panel displays a query in the 'mongosh' terminal:

```
>_MONGOSH
> db.book_metadata.find({
  avgRating: { $gt: 4 },
  formats: { $in: ["Digital"] }
})
< {
  _id: ObjectId('6a8d23960e5a8db208c7beee'),
  title: 'The Silent Algorithm',
  author: 'R. Menon',
  genre: 'Technology',
  publishedYear: 2022,
  formats: [
    'Hardcover',
    'Digital',
    'Audiobook'
  ],
  specs: {
    pages: 340,
    fileSize_MB: 4.2,
    drm: true,
    narrator: 'K. Rao',
    runtime_min: 610
  },
  avgRating: 4.6,
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
```

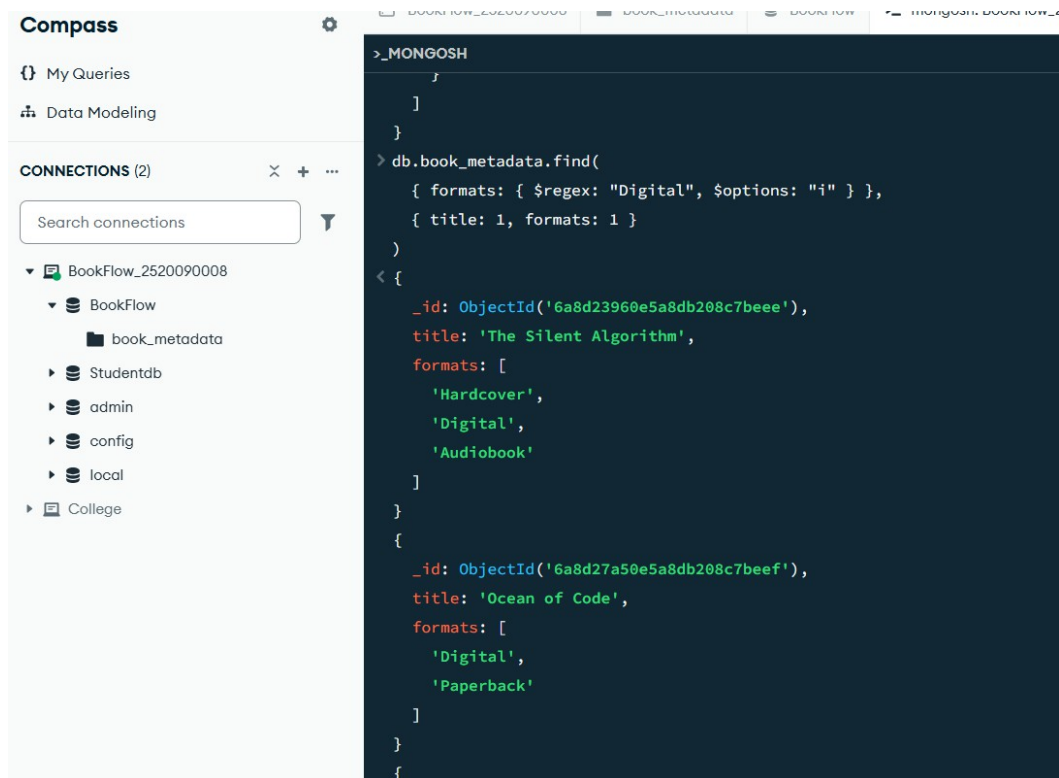


The screenshot shows the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' panel lists the database 'BookFlow_2520090008' and its collections 'BookFlow' and 'book_metadata'. The main panel displays a query in the 'mongosh' terminal:

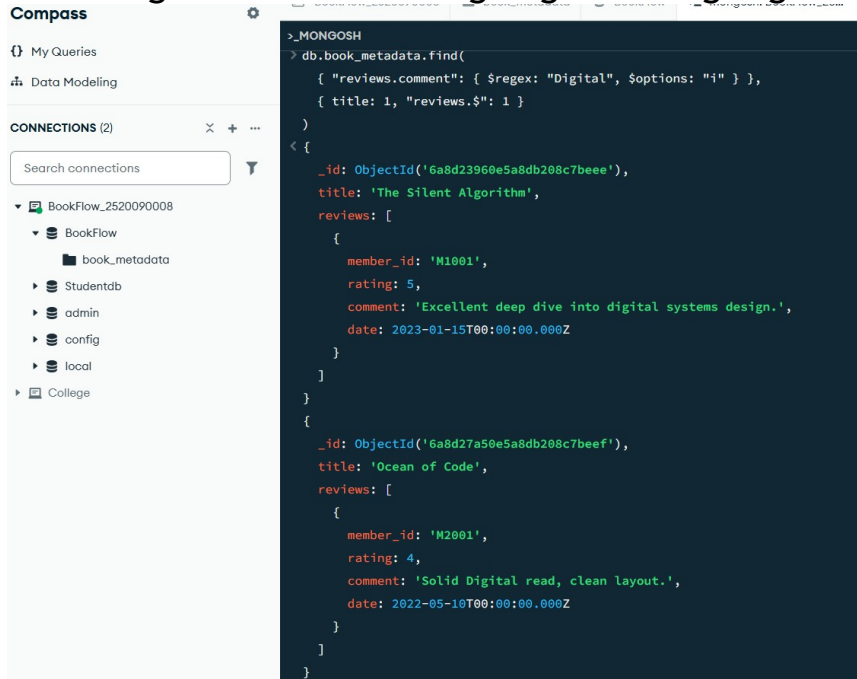
```
>_MONGOSH
{
  member_id: 'M1042',
  rating: 4,
  comment: 'Good but the Digital edition had formatting issues.',
  date: 2023-02-20T00:00:00.000Z
}
]
}
{
  _id: ObjectId('6a8d27a50e5a8db208c7beef'),
  title: 'Ocean of Code',
  author: 'A. Fernandes',
  genre: 'Technology',
  publishedYear: 2021,
  formats: [
    'Digital',
    'Paperback'
  ],
  specs: {
    pages: 210,
    fileSize_MB: 2.1,
    drm: false
  },
  avgRating: 4.2,
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.'
    }
  ]
}
```



Finding Digital Books Using Regular Expression



Searching Reviews Containing "Digital" Using Regex

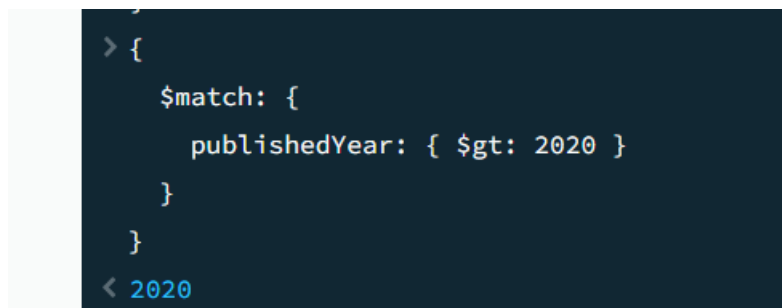


The screenshot shows the MongoDB Compass interface. On the left, the 'Connections' panel shows a connection to 'BookFlow_2520090008'. The 'Data Modeling' tab is active. The main panel shows a query in the MongoDB Shell:

```
>_MONGOSH
> db.book_metadata.find(
  { "reviews.comment": { $regex: "Digital", $options: "i" } },
  { title: 1, "reviews.$": 1 }
)
< {
  _id: ObjectId('6a8d23960e5a8db208c7beee'),
  title: 'The Silent Algorithm',
  reviews: [
    {
      member_id: 'M1001',
      rating: 5,
      comment: 'Excellent deep dive into digital systems design.',
      date: 2023-01-15T00:00:00.000Z
    }
  ]
}
{
  _id: ObjectId('6a8d27a50e5a8db208c7beef'),
  title: 'Ocean of Code',
  reviews: [
    {
      member_id: 'M2001',
      rating: 4,
      comment: 'Solid Digital read, clean layout.',
      date: 2022-05-10T00:00:00.000Z
    }
  ]
}
```

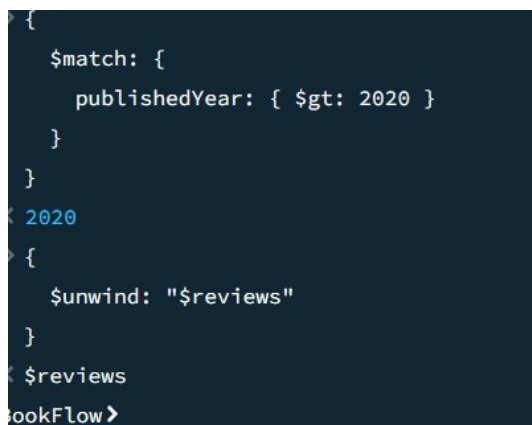
TASK - 3:

\$match — Books Published After 2020



```
> {
  $match: {
    publishedYear: { $gt: 2020 }
  }
}
< 2020
```

\$unwind — Expand Reviews



```
{
  $match: {
    publishedYear: { $gt: 2020 }
  }
}
< 2020
> {
  $unwind: "$reviews"
}
< $reviews
bookFlow>
```

Grouping Books and Calculating Average Rating Using \$group

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow_2520090008

BookFlow

book_metadata

Studentdb

admin

config

local

College

>_MONGOSH

```
db.book_metadata.aggregate([
  {
    $match: {
      publishedYear: { $gt: 2020 }
    }
  },
  {
    $unwind: "$reviews"
  },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  }
])
```

< {

```
  _id: 'Quantum Horizons',
  avgReviewRating: 5,
  totalReviews: 2,
  author: 'R. Menon'
}
```

{

```
  _id: 'The Silent Algorithm',
  avgReviewRating: 4.5,
  totalReviews: 2,
```

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow_2520090008

BookFlow

book_metadata

Studentdb

admin

config

local

College

{

```
  $group: {
    _id: "$title",
    avgReviewRating: { $avg: "$reviews.rating" },
    totalReviews: { $sum: 1 },
    author: { $first: "$author" }
  }
}
```

])

< {

```
  _id: 'Quantum Horizons',
  avgReviewRating: 5,
  totalReviews: 2,
  author: 'R. Menon'
}
```

{

```
  _id: 'The Silent Algorithm',
  avgReviewRating: 4.5,
  totalReviews: 2,
  author: 'R. Menon'
}
```

{

```
  _id: 'Ocean of Code',
  avgReviewRating: 4.5,
  totalReviews: 2,
  author: 'A. Fernandes'
}
```

}

BookFlow

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- BookFlow_2520090008
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
- College

```
>_MONGOSH
{
  _id: 0,
  title: "$_id",
  author: 1,
  avgReviewRating: {
    $round: ["$avgReviewRating", 2]
  },
  totalReviews: 1
}
})
< {
  totalReviews: 2,
  author: 'R. Menon',
  title: 'Quantum Horizons',
  avgReviewRating: 5
}
{
  totalReviews: 2,
  author: 'A. Fernandes',
  title: 'Ocean of Code',
  avgReviewRating: 4.5
}
{
  totalReviews: 2,
  author: 'R. Menon',
  title: 'The Silent Algorithm',
  avgReviewRating: 4.5
}
}
BookFlow>
```

Sorting Books by Average Rating Using \$sort

My Queries

Data Modeling

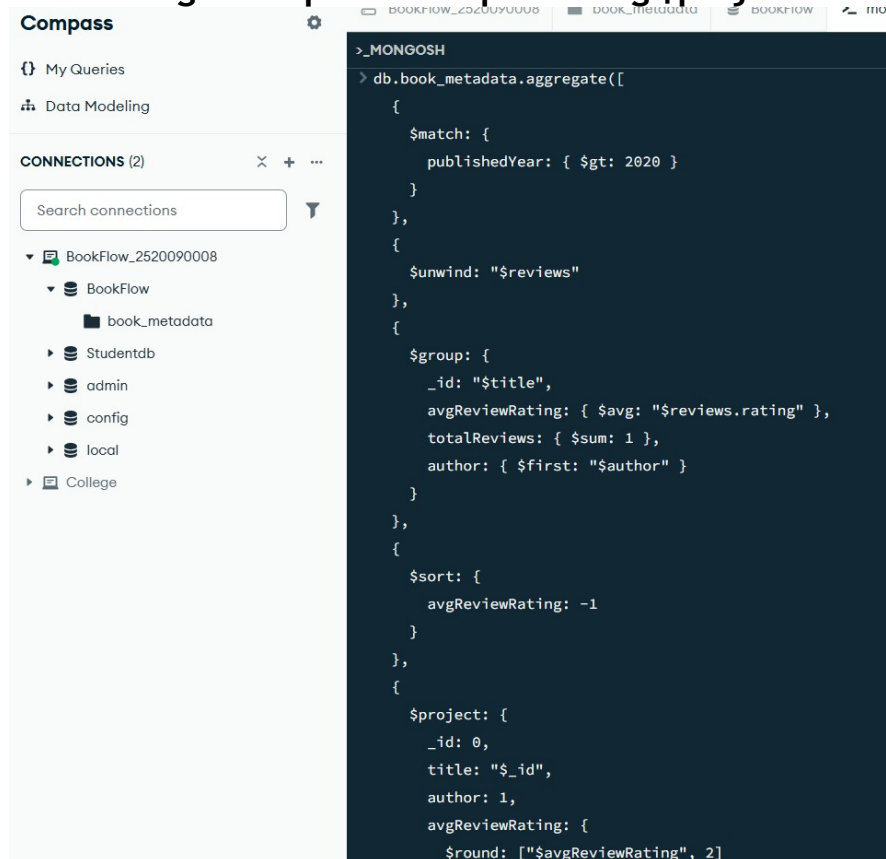
CONNECTIONS (2)

Search connections

- BookFlow_2520090008
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
- College

```
>_MONGOSH
author: { $first: "$author" }
}
})
< {
  _id: 'Quantum Horizons',
  avgReviewRating: 5,
  totalReviews: 2,
  author: 'R. Menon'
}
{
  _id: 'The Silent Algorithm',
  avgReviewRating: 4.5,
  totalReviews: 2,
  author: 'R. Menon'
}
{
  _id: 'Ocean of Code',
  avgReviewRating: 4.5,
  totalReviews: 2,
  author: 'A. Fernandes'
}
> {
  $sort: {
    avgReviewRating: -1
  }
}
< -1
BookFlow>
```

Formatting the Top Rated Report Using \$project

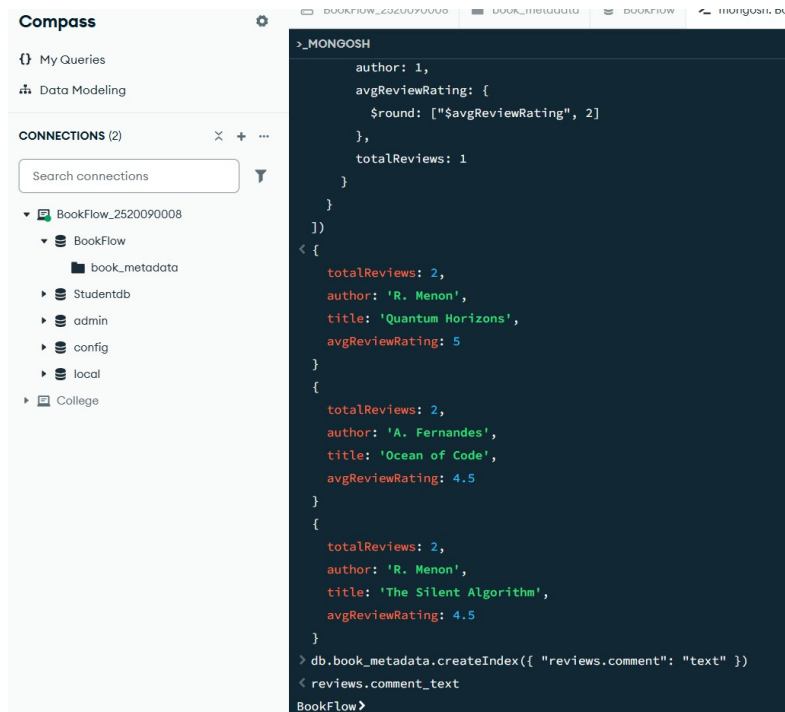


The screenshot displays the MongoDB Compass interface. On the left, the 'CONNECTIONS (2)' sidebar shows a tree view with 'BookFlow_2520090008' expanded, revealing 'BookFlow' and 'book_metadata' collections. The main area on the right is the MONGOSH console, which contains the following aggregation pipeline:

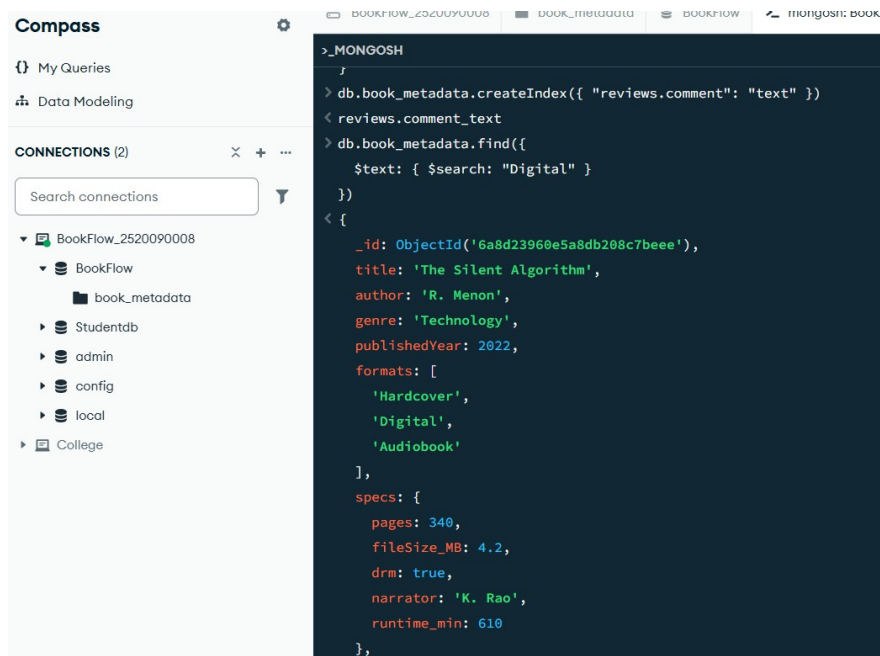
```
>_MONGOSH
> db.book_metadata.aggregate([
  {
    $match: {
      publishedYear: { $gt: 2020 }
    }
  },
  {
    $unwind: "$reviews"
  },
  {
    $group: {
      _id: "$title",
      avgReviewRating: { $avg: "$reviews.rating" },
      totalReviews: { $sum: 1 },
      author: { $first: "$author" }
    }
  },
  {
    $sort: {
      avgReviewRating: -1
    }
  },
  {
    $project: {
      _id: 0,
      title: "$_id",
      author: 1,
      avgReviewRating: {
        $round: ["$avgReviewRating", 2]
      }
    }
  }
])
```

Task 4: Indexing for Speed

1. Creating a Text Index on Review Comments



2. Searching Review Comments Using the Text Index



3. Verifying Text Index Usage Using .explain()

Compass

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

- BookFlow_2520090008
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
- College

```
>_MONGOSH
}
> db.book_metadata.find({
  $text: { $search: "Digital" }
}).explain("executionStats")
< {
  explainVersion: '1',
  queryPlanner: {
    namespace: 'BookFlow.book_metadata',
    parsedQuery: {
      '$text': {
        '$search': 'Digital',
        '$language': 'english',
        '$caseSensitive': false,
        '$diacriticSensitive': false
      }
    },
    indexFilterSet: false,
    queryHash: '4FB85230',
    planCacheShapeHash: '4FB85230',
    planCacheKey: 'F34F3876',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExploreReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'TEXT_MATCH',
      nss: 'BookFlow.book_metadata',
```

Compass

My Queries

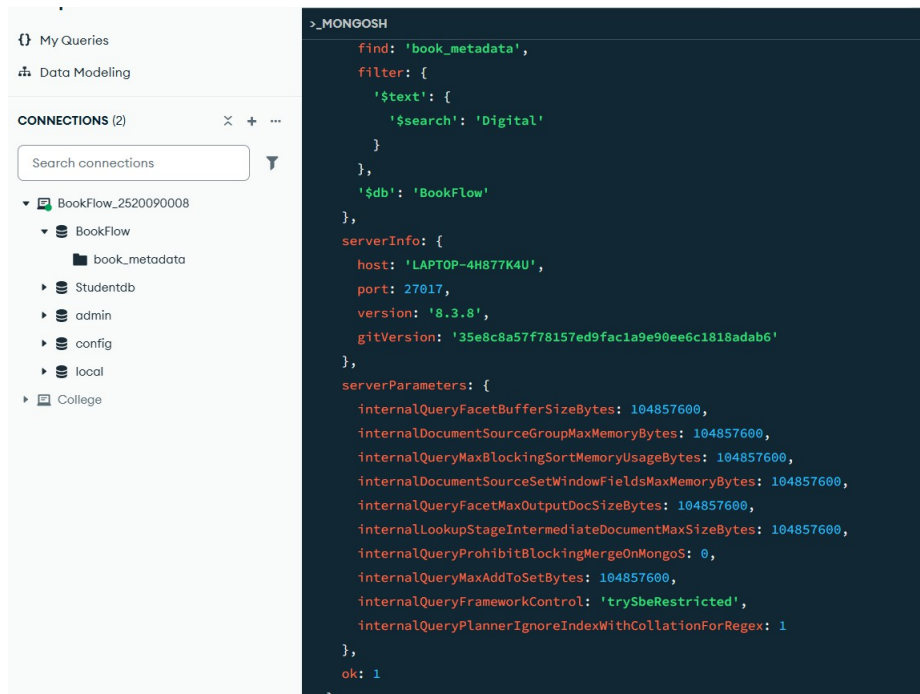
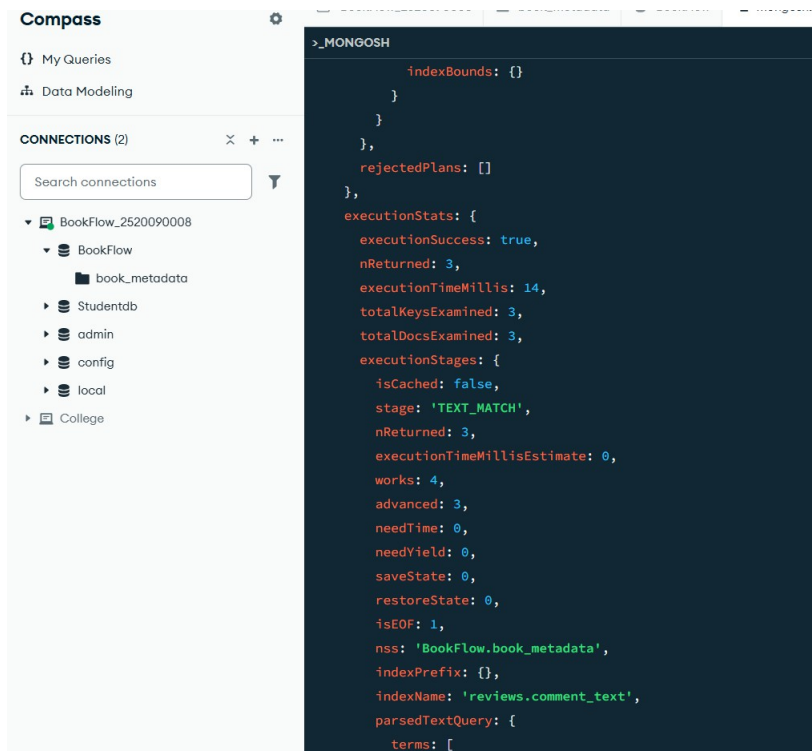
Data Modeling

CONNECTIONS (2)

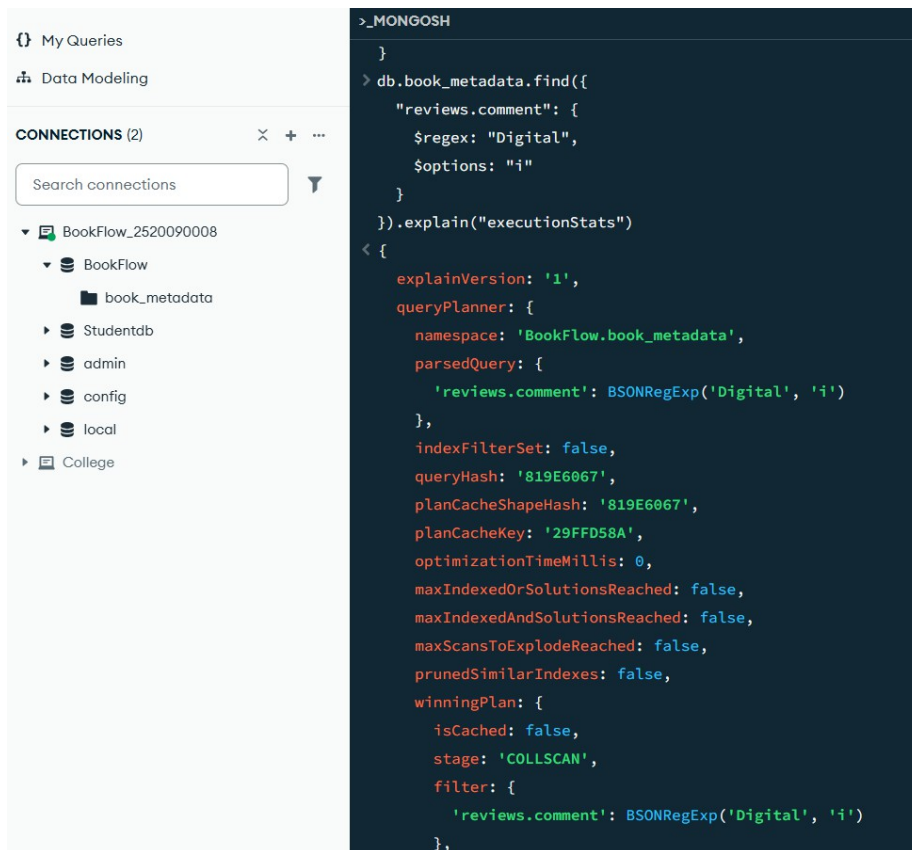
Search connections

- BookFlow_2520090008
 - BookFlow
 - book_metadata
 - Studentdb
 - admin
 - config
 - local
- College

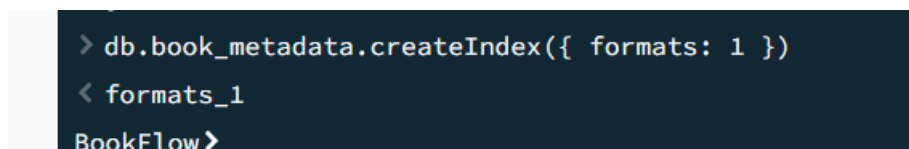
```
>_MONGOSH
  nss: 'BookFlow.book_metadata',
  indexPrefix: {},
  indexName: 'reviews.comment_text',
  parsedTextQuery: {
    terms: [
      'digit'
    ],
    negatedTerms: [],
    phrases: [],
    negatedPhrases: []
  },
  textIndexVersion: 3,
  inputStage: {
    stage: 'FETCH',
    nss: 'BookFlow.book_metadata',
    inputStage: {
      stage: 'IXSCAN',
      nss: 'BookFlow.book_metadata',
      keyPattern: {
        _fts: 'text',
        _ftsx: 1
      },
      indexName: 'reviews.comment_text',
      isMultiKey: true,
      isUnique: false,
      isSparse: false,
      isPartial: false,
      indexVersion: 2,
      direction: 'backward',
```



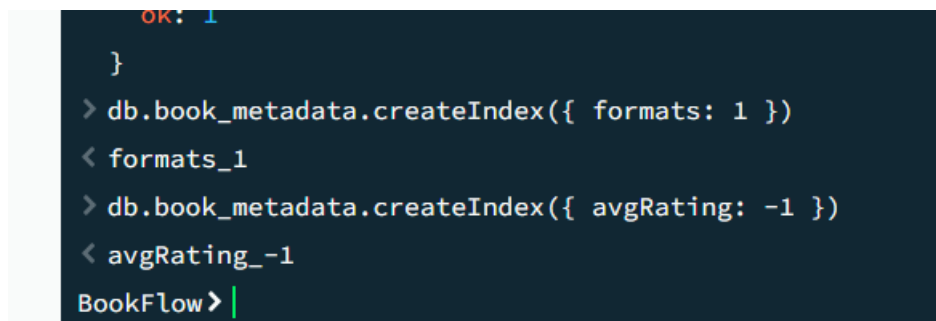
4. Comparing Regex Search Performance Using .explain()



5. Creating a Multikey Index on Book Formats



6. Creating an Index on Average Rating



7. Verifying Average Rating Index Using .explain()

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

▼ BookFlow_2520090008

▼ BookFlow

book_metadata

Studentdb

admin

config

local

College

>_MONGOSH

> db.book_metadata.find({
 avgRating: { \$gt: 4 }
}).explain("executionStats")
< {
 explainVersion: '1',
 queryPlanner: {
 namespace: 'BookFlow.book_metadata',
 parsedQuery: {
 avgRating: {
 '\$gt': 4
 }
 },
 },
 indexFilterSet: false,
 queryHash: 'E21FCB34',
 planCacheShapeHash: 'E21FCB34',
 planCacheKey: '6A4442CC',
 optimizationTimeMillis: 5,
 maxIndexedOrSolutionsReached: false,
 maxIndexedAndSolutionsReached: false,
 maxScansToExplodeReached: false,
 prunedSimilarIndexes: false,
 winningPlan: {
 isCached: false,
 stage: 'FETCH',
 nss: 'BookFlow.book_metadata',
 inputStage: {
 stage: 'IXSCAN',
 nss: 'BookFlow.book_metadata',

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

▼ BookFlow_2520090008

▼ BookFlow

book_metadata

Studentdb

admin

config

local

College

>_MONGOSH

ns: 'BookFlow.book_metadata',
 keyPattern: {
 avgRating: -1
 },
 indexName: 'avgRating_-1',
 isMultiKey: false,
 multiKeyPaths: {
 avgRating: []
 },
 isUnique: false,
 isSparse: false,
 isPartial: false,
 indexVersion: 2,
 direction: 'forward',
 indexBounds: {
 avgRating: [
 '[inf, 4)'
]
 }
},
 rejectedPlans: []
},
 executionStats: {
 executionSuccess: true,
 nReturned: 3,
 executionTimeMillis: 14,
 totalKeysExamined: 3,
 totalDocsExamined: 3,
 executionStages: {

My Queries

Data Modeling

CONNECTIONS (2)

Search connections

BookFlow_2520090008

BookFlow

book_metadata

Studentdb

admin

config

local

College

2_MONGO

```
find: 'book_metadata',
filter: {
  avgRating: {
    '$gt': 4
  }
},
'$db': 'BookFlow'
},
serverInfo: {
  host: 'LAPTOP-4H877K4U',
  port: 27017,
  version: '8.3.8',
  gitVersion: '35e8c8a57f78157ed9fac1a9e90ee6c1818adab6'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
```